

Coarse-to-Fine Saliency Fusion for Interdependent Video Moment Retrieval and Highlight Detection

by

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ABSTRACT

Video moment retrieval (MR) and highlight detection (HD) conditioned on natural language queries are two highly related tasks, aiming to localize relevant temporal moments in videos and predict clip-level highlight scores. Although recent DETR-based methods jointly model MR and HD by attaching separate task heads after multimodal interaction, they often have limited performance to make full use of the interrelationship between the two tasks. In this thesis, we propose OCS-DETR, a task-reciprocal DETR framework with coarse-to-fine saliency fusion to strengthen mutual feedback between MR and HD.

First, on the input side, we introduce a gated Optimal Transport Aligner (OTAligner) to perform bidirectional soft alignment between video and text tokens in a shared latent space, and we further enhance cross-modal fusion by applying Virtual Refinement Fuser (VRFuser) prior to the transformer encoder. Then, inside the transformer, we design Saliency-Guided Refinement (SGR) to estimate clip saliency and reweight visual tokens for a second-stage encoding, allowing highlight cues to suppress query-irrelevant information and improve retrieval quality. On the output side, we propose Coarse-to-Fine Moment Attention Fusion (CMAF), that converts top-K MR proposals into a differentiable soft-window prior and cosine-based temporal attention to refine highlight prediction, enabling MR-to-HD feedback. We also adopt focal loss to address severe foreground-background imbalance and remove the expensive negative-pair forward pass used in prior baselines, improving training efficiency.

Experiments on QVHighlights [3] and TVSum [18] demonstrate consistent gains from each component, achieving strong joint MR/HD performance. Overall, OCS-DETR provides an effective and efficient solution for reciprocal moment retrieval and highlight detection.

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1 Introduction

Imagine searching for a single moment of “scoring a goal” inside a two-hour basketball replay, or trying to locate the exact moment a YouTuber folds egg in a 30-minute cooking vlog. With the development of digital devices and the rapid growth of online platforms, the quantity and diversity of video content have increased dramatically. Manually scrubbing through the full timeline makes users feel inconvenienced and frustrating. As a result, users increasingly expect systems that can quickly locate the desired moments in long videos and efficiently browse the most informative parts instead of watching the full timeline. Two video understanding tasks address this demand: **video moment retrieval (MR)**, which localizes the temporal segment relevant to a natural language query, and **highlight detection (HD)**, which assigns clip-level saliency scores indicating which parts are most worth watching under the same query.

Although these tasks have a shared goal, which is understanding video content with respect to user intent, MR and HD remain challenging. First, videos are often long and redundant, and the content can be visually complex. Second, matching text tokens with video clips is noisy, even Transformer encoders [1] may assign high relevance to weakly related clips, which hurts alignment and makes saliency unstable. Third, MR and HD optimize different goals: MR cares more about accurate boundaries, while HD cares more about ranking contrast across clips.

Recent works have begun to jointly model MR and HD using DETR-style architectures [2], where a fixed set of learned queries decodes multiple candidate temporal spans and associated confidence scores. Moment-DETR demonstrates that MR and HD can be learned together in an end-to-end set prediction framework, enabling multi-moment localization and clip-wise saliency estimation in a unified pipeline [3]. Building on this direction, QD-DETR [4] introduces query-dependent video representations by injecting query information early and further employing negative video–query pairs to encourage low saliency on irrelevant pairs. These DETR-based joint frameworks achieve strong performance by improving multimodal encoding and interaction.

However, many joint methods still use two separate heads after a shared encoder, which means MR and HD do not really help each other during prediction. This design ignores the reciprocal relationship between the tasks. We think this is a missed opportunity because saliency cues can guide localization, and good spans can also provide a time prior for saliency. If HD can highlight which clips are truly important for the query, it can help MR suppress query-irrelevant content and sharpen temporal localization. Inversely, if MR can provide reliable top candidate spans, it can serve as a structured temporal prior that stabilizes and refines HD scoring. In other words, MR and HD are not merely co-existing objectives; they can be mutual refinement signals within a single model.

To better exploit this reciprocity while strengthening cross-modal grounding, we propose OCS-DETR, a DETR-based framework featuring Optimal-Transport alignment and Coarse-to-Fine Saliency Fusion for interdependent MR and HD. OCS-DETR is designed around three stages:

(1) Input-side cross-modal alignment and fusion. We introduce an Optimal Transport Aligner (OTAligner) that performs token-level soft alignment between video clips and text tokens in a shared latent space. Using an entropy-regularized OT solver (Sinkhorn-style computation) [5], OTAligner produces a bidirectional alignment signal that is injected back into the original features via a learnable gated residual, improving semantic correspondence before the main Transformer reasoning. After OT-based alignment, we further apply VRFuser, a cross-modal interaction module placed before the DETR-style Transformer, to strengthen query-conditioned fusion at the earliest stage. VRFuser is adapted from the Context–Query Attention (CQA) design in VSLNet [7], which implements bidirectional context-to-query and query-to-context attentions—an idea widely used in reading comprehension models such as BiDAF and QANet [8] [9].

(2) In-Transformer refinement with HD to MR feedback. Inside the Transformer encoder, we introduce Saliency-Guided Refinement (SGR). SGR first estimates clip-wise saliency and then uses the resulting temporal weights to amplify potentially highlight-relevant tokens, followed by a refinement encoding step. This mechanism allows highlight cues to explicitly suppress query-irrelevant information in the video representation, improving the quality of features used for retrieval.

(3) Output-side refinement with MR to HD feedback. After decoding candidate moment spans, we propose Coarse-to-Fine Moment Attention Fusion (CMAF) to feed retrieval outcomes back to highlight prediction. CMAF converts the top-K retrieved spans into a differentiable soft-window prior over time, combines it with similarity-driven temporal attention, and writes the fused information back to refine the saliency distribution. This provides a structured “where to look” prior derived from MR, improving HD ranking stability.

Finally, OCS-DETR adopts Focal Loss [6] to mitigate severe foreground–background imbalance in DETR-style query classification, and removes the extra negative-pair forward pass commonly used in query-dependent baselines such as QD-DETR [4], improving training efficiency while relying on architectural reciprocity (SGR/CMAF) to achieve stronger joint behavior.

To summarize, our work has these main points:

- Base on QD-DETR [4], we build a DETR-style joint MR/HD model that allows HD help MR and MR help HD feedback (SGR and CMAF).

- We introduce a gated OTAligner to enforce stronger token-level cross-modal alignment before Transformer reasoning, improving semantic faithfulness of interaction.
- We further propose VRFuser, a lightweight pre-Transformer cross-modal interaction module adapted from Context–Query Attention (CQA) in VSLNet [7] to strengthen query-conditioned fusion at the earliest stage.
- We design a coarse-to-fine saliency fusion mechanism that uses retrieved moment proposals as a differentiable temporal prior to refine highlight prediction.
- We improve training robustness and efficiency via Focal Loss and by removing an expensive negative-pair forward strategy used in prior query-dependent DETR baselines.

2 Related Work

2.1 Query-based Moment Retrieval and Highlight Detection

Video moment retrieval (MR) aims to localize the start and end time of the video segment that semantically matches a natural language query [10]. Existing MR methods are commonly grouped into proposal-based and proposal-free paradigms. Proposal-based approaches first generate candidate moments (e.g., sliding windows or dense temporal proposals) and then rank them by query–video similarity; 2D temporal maps further model start–end combinations explicitly to improve boundary reasoning [10] [11]. While effective, proposal-based pipelines often introduce extra computation and depend on proposal quality. In contrast, proposal-free methods directly predict span boundaries or moment distributions without enumerating a large candidate set, which is typically more efficient and easier to train end-to-end.

Highlight detection (HD) scores each clip’s saliency. Traditional HD is often query-agnostic, but query-conditioned HD better reflects user intent by allowing highlights to change under different queries [3]. Recent works also show that strong vision–language representations can be highly competitive for highlight scoring and ranking under query conditioning [14].

2.2 DETR-style Joint Modeling of MR and HD

Joint learning of MR and HD became practical after unified benchmarks such as QVHighlights and DETR-style architectures were introduced [3]. Moment-DETR pioneers the application of DETR-style set prediction for joint MR/HD on QVHighlights [3]. A representative line of work applies DETR-style set prediction to temporal grounding, decoding a fixed number of candidate moments via learned queries and training with matching-based objectives [2] [3]. Building upon this paradigm, QD-DETR [4] strengthens query-dependent video representations by injecting query information early and employing additional training strategies to discourage irrelevant saliency. These joint frameworks typically attach two task-specific heads after a shared multimodal encoder, achieving strong results by improving feature interaction.

However, such designs often treat MR and HD as parallel outputs, and may underexploit their reciprocity. TR-DETR [12] explicitly argues that HD cues can help suppress query-irrelevant clips for MR, while MR localization can provide temporal priors to refine HD, proposing a task-cooperation mechanism to realize HD help MR and MR help HD interactions. Our OCS-DETR

follows this motivation, but focuses on strengthening token-level cross-modal alignment and building a coarse-to-fine saliency fusion loop to realize task reciprocity within a DETR-style pipeline.

2.3 Cross-modal Interaction and Multi-modal Alignment

Cross-modal interaction is central to MR/HD, since the system must connect query tokens with video clips. A widely adopted interaction primitive is Context–Query Attention (CQA), which computes a similarity matrix between context and query, derives context-to-query and query-to-context attentions, and fuses them through concatenation and projection. This bidirectional attention formulation is well established in reading comprehension models such as BiDAF and QANet [8] [9], and has been adapted to video-language grounding in VSLNet [7] for fine-grained video–text interaction.

Beyond attention-based fusion, another direction is to impose explicit alignment constraints between modalities. Contrastive learning has been widely used to align video and text representations in a shared space, improving retrieval-oriented objectives [13]. More structured alignment can be formulated with Optimal Transport (OT), where a transport plan provides soft token-level matching under a principled cost, and Sinkhorn-style solvers make OT practical for integration into deep models [5]. These ideas motivate OCS-DETR’s gated OTAligner for stronger pre-Transformer semantic alignment.

3 Methodology

Overview

The overall architecture of OCS-DETR shown in Figure 1, is built upon the DETR-style joint framework for moment retrieval (MR) and highlight detection (HD) [2] [4]. To ensure a fair comparison, we inherit the same frozen feature extraction pipeline from QD-DETR [4]. Therefore, performance gains mainly come from our OT-based alignment and fusion / interdependency design, rather than stronger backbones.

OCS-DETR contains three core parts: (1) Feature Extraction Part, (2) Alignment Part, (3) Interdependency Part (HD help MR, MR help HD).

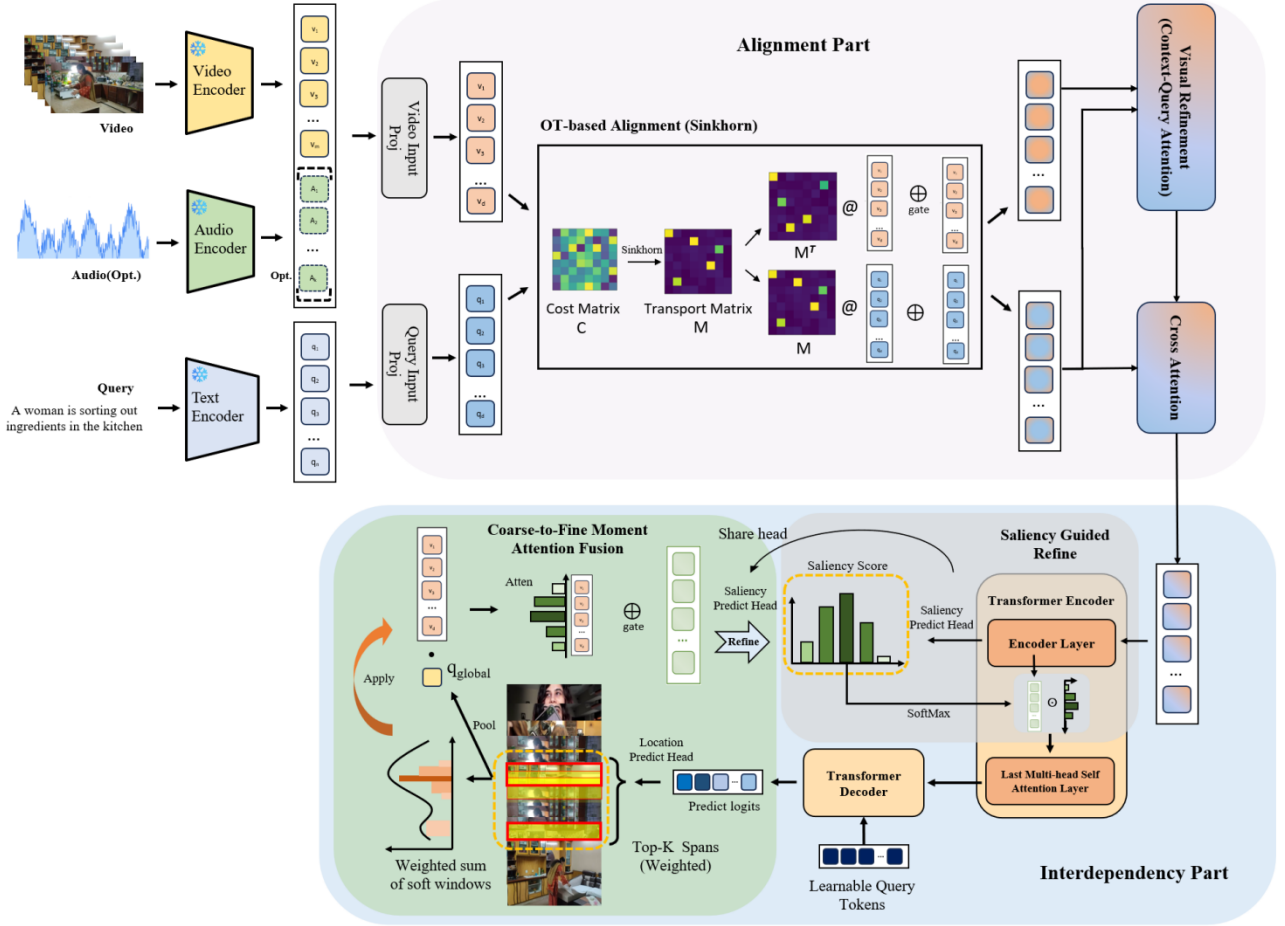


Figure 1: The proposed OCS-DETR consists of several key stages to jointly model moment retrieval (MR) and highlight detection (HD). First, we use pre-trained Video and Text Encoders (with an optional Audio Encoder) to extract modality-specific features, and project them into a shared embedding space. Next, in the Alignment Part, we perform OT-based alignment (Sinkhorn) by building a cost matrix and solving a transport matrix, then using a learnable gate to fuse the aligned features in a stable residual manner. After alignment, we apply visual refinement via context–query attention and further cross-attention to strengthen token-level video–text interaction. Finally, in the Interdependency Part, a DETR-style Transformer encoder–decoder with learnable query tokens predicts moment locations, while Saliency-Guided Refinement (SGR) suppresses non-highlight clips by re-encoding features under saliency guidance. In addition, Coarse-to-Fine Moment Attention Fusion (CMAF) uses top-K MR proposals as soft temporal priors, so MR help HD by refining highlight scores with moment-focused attention.

3.1 Feature Extraction

We follow the QD-DETR feature pipeline [4]. Given a video segmented into L clips and a query with N tokens:

Visual features. For each clip, we concatenate CLIP visual semantics with SlowFast motion dynamics:

$$\mathbf{F}_v = [\mathbf{F}_{CLIP}; \mathbf{F}_{SlowFast}] \in \mathbb{R}^{L \times d_v}$$

This captures both high-level semantics and temporal motion cues [15] [16].

Audio features (optional). We optionally extract audio features using PANN and concatenate them to the visual stream:

$$\mathbf{F}_{va} = [\mathbf{F}_v; \mathbf{F}_a] \in \mathbb{R}^{L \times (d_v + d_a)}$$

[17]

Text features. We use the CLIP text encoder to obtain token-level query features:

[15]

Projection to shared dimension. We map modalities to a common hidden size d using MLP stacks:

$$\mathbf{V} = MLP_v(\mathbf{F}_{va}) \in \mathbb{R}^{L \times d}, \quad \mathbf{T} = MLP_t(\mathbf{F}_t) \in \mathbb{R}^{N \times d}$$

3.2 Feature Alignment

3.2.1 OT-Based Alignment (Sinkhorn)

A major challenge in MR and HD is token-level mismatch between video clips and query words. We introduce an optimal transport (OT) alignment module to softly align $\mathbf{V}$ and $\mathbf{T}$ before deep fusion.

Cost matrix. We first compute a cosine-based cost between projected features:

$$\mathbf{C}_{i,j} = 1 - \cos(\mathbf{W}\mathbf{V}_i, \mathbf{W}\mathbf{T}_j), \quad \mathbf{C} \in \mathbb{R}^{L \times N}$$

Sinkhorn transport plan. We solve an entropic-regularized OT plan with Sinkhorn iterations:

$$\mathbf{M} = \text{Sinkhorn}(\mathbf{C}; \epsilon, K), \quad \mathbf{M} \in \mathbb{R}^{L \times N}$$

where ϵ is the regularization strength and K is the number of iterations [5].

Bidirectional soft alignment. Using $\mathbf{M}$, we obtain aligned features:

$$\tilde{\mathbf{V}} = \mathbf{M}\mathbf{T} \in \mathbb{R}^{L \times d}, \quad \tilde{\mathbf{T}} = \mathbf{M}^\top \mathbf{V} \in \mathbb{R}^{N \times d}$$

We initially tried to directly replace the original features with OT-aligned features (using $\tilde{\mathbf{V}}$ as the input of fusion). However, we observed performance drops, likely because early OT plans are noisy when encoders are frozen. To address this, we introduce a learnable gate γ initialized to 0, so the model starts exactly from the stable QD-DETR [4] behavior and gradually increases the contribution of OT alignment only when it becomes reliable.

Gated residual fusion. We inject alignment with a learnable scalar gate γ (initialized to 0):

$$\mathbf{V}' = \mathbf{V} + \gamma \tilde{\mathbf{V}}, \quad \mathbf{T}' = \mathbf{T} + \gamma \tilde{\mathbf{T}}$$

This design lets the model start from the stable baseline behavior and gradually learn how much OT alignment it needs.

3.2.2 Visual Refinement Fuser

After OT alignment, we refine video features with a two-stage context–query attention (CQA) fusion module (VRFuser) (as shown in Figure 2), inspired by span-based QA fusion used in VSLNet [7] and related bidirectional attention formulations [8] [9].

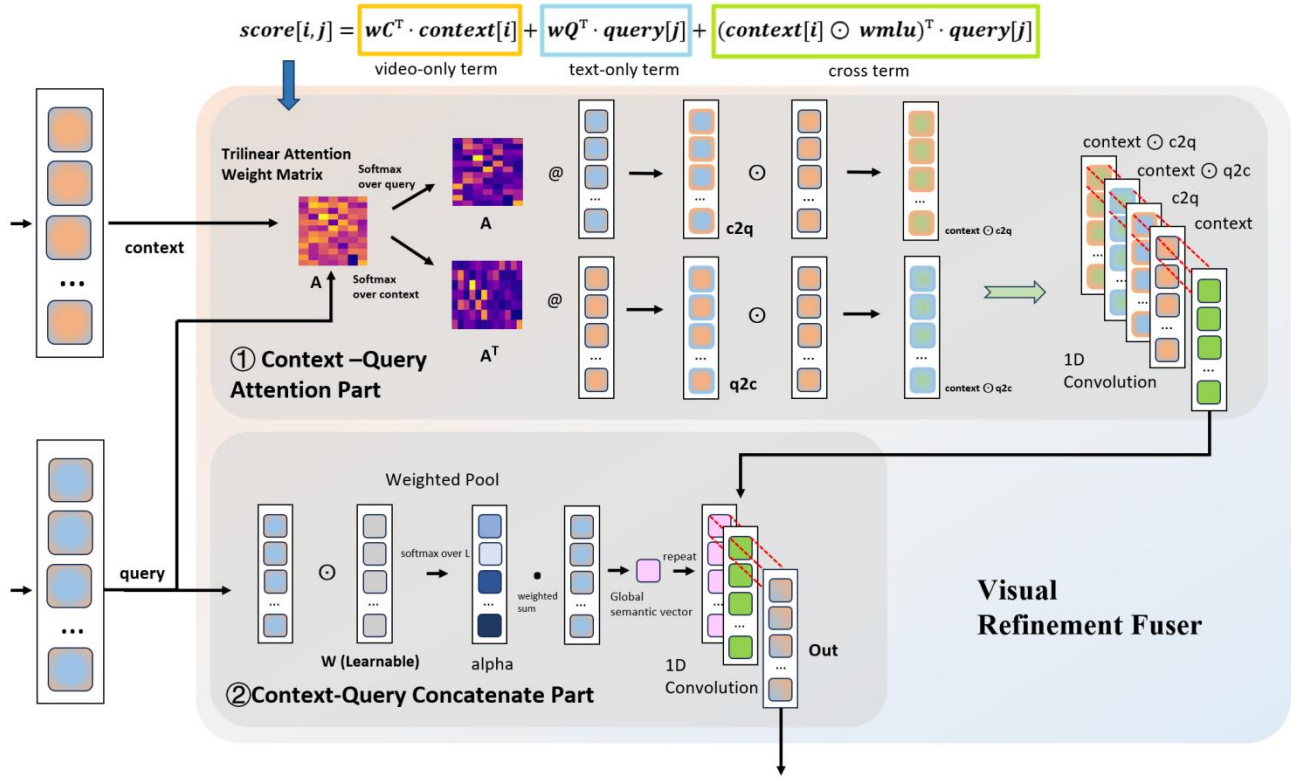


Figure 2: VRFuser illustrates a two-stage context–query attention (CQA) fusion for video–text interaction. In the first stage, it computes a trilinear attention score matrix between video (context) and text (query), and derives $c2q$ and $q2c$ attentions via softmax along query and context dimensions. The module then concatenates multi-branch interaction features (e.g., context, $c2q$, and element-wise products) and applies 1D convolution for feature projection. In the second stage, a learnable weighted pooling summarizes the query into a global semantic vector, which is broadcast and concatenated back to each time step, followed by another 1D convolution to produce the final fused output. This design first aligns tokens, then injects global query meaning to reduce semantic drift.

Context–Query Attention

Let context be $\mathbf{V}'$ and query be $\mathbf{T}'$. We compute a trilinear similarity:

$$S_{i,j} = \mathbf{w}_c^\top \mathbf{V}'_i + \mathbf{w}_q^\top \mathbf{T}'_j + ((\mathbf{V}'_i \odot \mathbf{w}_{mlu})^\top \mathbf{T}'_j)$$

We obtain attention distributions:

$$\mathbf{A}^{c2q} = \text{Softmax}_j(S), \quad \mathbf{A}^{q2c} = \text{Softmax}_i(S)$$

Then:

$$\begin{aligned} c2q_i &= \sum_j A_{i,j}^{c2q} \mathbf{T}'_j, \\ q2c_i &= \sum_j \left(\sum_k A_{i,k}^{c2q} A_{k,j}^{q2c} \right) \mathbf{V}'_j \end{aligned}$$

We form a multi-branch fusion vector and project it with 1D convolution (equivalent to position-wise FFN but efficient for sequences):

$$\mathbf{H}_i = \text{Conv1D}([\mathbf{V}'_i; c2q_i; \mathbf{V}'_i \odot c2q_i; \mathbf{V}'_i \odot q2c_i])$$

In our early experience, we use a single-direction query-to-video attention (only $c2q$) improved coarse relevance but often missed fine-grained temporal cues under long queries. This motivated us to adopt the bidirectional $c2q$ and $q2c$ formulation for richer interaction (Like original VSLNet [7] do). We further found that purely token-level fusion can drift when the query contains multiple entities or attributes. Therefore, we add a lightweight global query vector $\mathbf{q}_g$ via learnable pooling and inject it back to every time step for consistent conditioning.

Context–Query Concatenate (global query conditioning)

We summarize the query by a learnable weighted pooling:

$$\alpha_j = \text{Softmax}_j(\mathbf{w}^\top \mathbf{T}'_j), \quad \mathbf{q}_g = \sum_j \alpha_j \mathbf{T}'_j$$

Broadcast $\mathbf{q}_g$ and concatenate to each time step:

$$\hat{\mathbf{V}}_i = \text{Conv1D}([\mathbf{H}_i; \mathbf{q}_g])$$

This “token alignment + global semantic injection” structure reduces semantic drift and improves discrimination under complex queries. The Transformer can start from cleaner cross-modal features, which stabilizes training and improves both moment retrieval and highlight detection.

3.3 DETR-style Joint Modeling with Interdependency

We adopt the DETR paradigm for set prediction with learnable query tokens [2], and use Transformer attention for temporal reasoning [1] [4].

We concatenate refined video and text streams:

$$\mathbf{X} = [\hat{\mathbf{V}}; \mathbf{T}'] \in \mathbb{R}^{(L+N) \times d}$$

and feed them into the Transformer encoder–decoder to predict moment spans and clip saliency.

3.3.1 Saliency-Guided Refinement (SGR) (HD help MR)

Here, we first attempted hard masking by removing low-saliency clips before decoding, but this sometimes hurt recall when saliency was imperfect. We therefore use a soft, learnable re-weighting with residual write-back, which suppresses noise while preserving recoverable evidence.

We first estimate saliency scores from encoder features and use them to suppress query-irrelevant clips before decoding:

$$s_t = \frac{1}{\sqrt{d}} \sum_k (\mathbf{w}_s \mathbf{m}_t)_k, \quad t = 1, \dots, L$$

where $\mathbf{m}_t$ is the encoder memory for the t -th clip.

We apply masked softmax to obtain weights p_t and inject saliency-weighted features back via a residual refinement:

$$\mathbf{m}'_t = \mathbf{m}_t + p_t \cdot \mathbf{x}_t$$

followed by re-encoding. This makes MR decoding focus on salient and query-relevant clips (HD help MR).

3.3.2 Coarse-to-Fine Moment Attention Fusion (CMAF) (MR help HD)

To exploit task reciprocity, we use MR predictions as soft temporal priors to refine HD.

Top-K proposal prior. From DETR queries, we take top-K predictions based on foreground scores and convert predicted spans to $(start, end)$. We build differentiable “soft windows”:

$$\omega_t^{(j)} = \sigma\left(\frac{t - s_j}{\tau}\right) \cdot \sigma\left(\frac{e_j - t}{\tau}\right), \quad \sum_t \omega_t^{(j)} = 1$$

Using only the top-K predicted spans as a prior was sometimes brittle when early MR predictions were slightly shifted. To make the guidance more robust, we combine the span prior $\bar{\mathbf{w}}$ with a global similarity-based attention (cosine) in log space, so HD receives both “where MR looks” and “what is globally similar”.

Proposal fusion. Let $a_j = \text{Softmax}(score_j / \tau_p)$. We fuse a moment-focused global vector:

$$\mathbf{g} = \sum_{j=1}^K a_j \left(\sum_t \omega_t^{(j)} \mathbf{v}_t \right), \quad \bar{\mathbf{w}}_t = \sum_{j=1}^K a_j \omega_t^{(j)}$$

Cosine time attention + gated write-back. We compute cosine attention over the full timeline and combine it with the prior $\bar{\mathbf{w}}$:

$$\beta_t = \text{Softmax}(\log(\text{Softmax}(\cos(\mathbf{g}, \mathbf{v}_t))) + \log(\bar{\omega}_t))$$

We aggregate a value vector $\mathbf{u} = \sum_t \beta_t \mathbf{v}_t$ and update memory by a gated residual:

$$\mathbf{m}_t^* = \mathbf{m}_t + \alpha \mathbf{u}$$

Finally, refined highlight scores are produced from $\mathbf{m}^*$. This explicitly enforces MR help HD.

3.4 Prediction Heads and Objective Loss

Set-based moment prediction. Following the DETR-style set prediction paradigm, the Transformer decoder takes Q learnable query tokens and outputs a set of query embeddings $\{h_q\}_{q=1}^Q$ for moment retrieval [2] [4]. Each query q predicts (i) a foreground probability $\hat{y}_q$ and (ii) a normalized temporal span $\hat{m}_q$. A bipartite matching (Hungarian) assigns at most one predicted query to the ground-truth moment, and all other unmatched queries are treated as background—this is exactly why classification becomes extremely imbalanced in MR [2] [4].

3.4.1 Moment prediction head and loss

Span regression. For each query embedding h_q , a small MLP predicts a normalized center–width span:

$$\hat{m}_q = (\hat{c}_q, \hat{w}_q) \in [0, 1]^2$$

which can be converted to start–end $(\hat{s}_q, \hat{e}_q)$ for evaluation. We use the same regression losses as QD-DETR [4]:

$$\mathbf{L}_{L1} = \|\mathbf{m} - \hat{\mathbf{m}}\|_1, \quad \mathbf{L}_{gIoU} = 1 - \text{tIoU}(\mathbf{m}, \hat{\mathbf{m}})$$

where $\mathbf{m}$ is the matched ground-truth moment and tIoU is temporal IoU [4].

Moment classification (CE change to Focal). Different from QD-DETR, we replace the cross-entropy classification term with sigmoid **focal loss** to reduce domination by easy background queries (the majority) and emphasize hard negatives [6]. The MR loss becomes:

$$\mathbf{L}_{mr} = \lambda_{L1} \mathbf{L}_{L1} + \lambda_{gIoU} \mathbf{L}_{gIoU} + \lambda_{Focal} \mathbf{L}_{Focal}$$

The focal loss is computed over all decoder queries:

$$\mathbf{L}_{Focal} = - \sum_{q=1}^Q [\alpha (1 - p_q)^\gamma y_q \log(p_q) + (1 - \alpha) p_q^\gamma (1 - y_q) \log(1 - p_q)]$$

With

$$p_q = \begin{cases} \hat{y}_q, & y_q = 1 \\ 1 - \hat{y}_q, & y_q = 0 \end{cases}$$

Here $y_q \in \{0,1\}$ is the matched label (1 is foreground, 0 is background). By this way, we typically set $\alpha = 0.25$ and $\gamma = 2.0$ in experience, where the modulating factor $(1-p_q)^\gamma$ suppresses easy negatives and strengthens gradients for hard samples [6].

3.4.2 Highlight prediction head and loss

Saliency prediction. We keep the saliency supervision consistent with QD-DETR [4] for fair comparison: a saliency head outputs clip-level scores $S(x)$ for clips x , and supervision is given by ranking/margin constraints plus a contrastive ranking term. The highlight loss is:

$$L_{hl} = \lambda_{margin} L_{margin} + \lambda_{cont} L_{cont}$$

Margin loss. For a pair of clips (x^{high}, x^{low}) where x^{high} should be ranked higher than x^{low} , we enforce:

$$L_{margin} = \max(0, \Delta + S(x^{low}) - S(x^{high}))$$

so high-rank clips must score at least Δ higher than low-rank ones.

Contrastive rank loss. To build separable saliency levels across ranks, we use a temperature-scaled contrastive term over rank groups:

$$L_{cont} = - \sum_{r=1}^R \log \frac{\sum_{x \in X_r^{pos}} \exp\left(\frac{S(x)}{\tau}\right)}{\sum_{x \in (X_r^{pos} \cup X_r^{neg})} \exp\left(\frac{S(x)}{\tau}\right)}$$

where X_r^{pos} and X_r^{neg} denote positive/negative clip sets at rank level r , and τ controls whether the separation is soft or sharp.

3.4.3 Overall training objective

Putting MR and HL together, the final objective is:

$$L_{total}^{OCS} = L_{mr}^{(with focal)} + L_{hl}^{(margin+rank)}$$

MR uses L1 + gIoU + Focal, and HL uses margin + contrastive/rank. Compared with QD-DETR, the key change is CE help Focal in MR classification to address extreme fg/bg imbalance and improve matching stability [4] [6].

4 Experiments

4.1 Datasets

We evaluate the effectiveness of our proposed model on two benchmarks: QVHighlights [3] for joint moment retrieval (MR) and highlight detection (HD), and TVSum [18] for highlight detection.

4.1.1 QVHighlights

QVHighlights is a large-scale benchmark designed for joint evaluation of Moment Retrieval (MR) and Highlight Detection (HD). It contains 10,148 diverse YouTube videos (up to 150 seconds) with 10,310 annotations. Each annotation includes a free-form query, one or more relevant temporal moments (averaging 1.8 per query), and clip-level saliency scores. By supporting both MR and HD within a unified framework, QVHighlights enables comprehensive evaluation of joint modeling approaches. Test-set evaluation is conducted on an official server, and we follow the standard 70%/15%/15% split for training, validation, and testing.

4.1.2 TVSum

TVSum is a benchmark dataset for Highlight Detection, consisting of videos from 10 categories with five videos per category. Each video is annotated with segment-level importance scores from multiple users, enabling supervised learning of highlight ranking. Unlike MR datasets, TVSum does not involve textual queries and focuses solely on visual saliency modeling. We follow the standard 80%/20% train–test split in our experiments.

4.2 Evaluation Metrics

We follow standard evaluation protocols used in prior works on moment retrieval and highlight detection [3] [4] [18]. For moment retrieval, we report Recall@1 ($R@1$) at IoU thresholds of 0.5 and 0.7, as well as mean Average Precision (mAP) at IoU thresholds of 0.5 and 0.75. Following Lei et al. [3], mAP is computed by uniformly sampling 10 IoU thresholds from [0.5, 0.95] and averaging the results, referred to as Avg. mAP. For highlight detection, we evaluate performance using mAP and HIT@1, where HIT@1 measures whether the top-ranked clip overlaps with ground-truth highlights.

4.3 Experimental Results

4.3.1 Results on QVHighlights Dataset

Method	Src	Moment Retrieval					Highlight Detection	
		R1		mAP		Avg.	$\geq$ Very Good	
		@0.5	@0.7	@0.5	@0.75		mAP	HIT@1
BeautyThumb	V	-	-	-	-	-	14.36	20.88
DVSE	V	-	-	-	-	-	18.75	21.79
MCN	V	11.41	2.72	24.94	8.22	10.67	-	-
CAL	V	25.49	11.54	23.40	7.65	9.89	-	-
XML	V	41.83	30.35	44.63	31.73	32.14	34.49	55.25
XML+	V	46.69	33.46	47.89	34.89	34.90	35.38	55.06
Moment-DETR	V	52.89	33.02	54.80	29.80	35.09	35.79	55.60
QD-DETR*	V	62.40	44.98	62.52	39.88	39.86	38.94	62.40
UVCOM	V	<u>63.35</u>	<u>47.47</u>	<u>63.37</u>	<u>42.77</u>	43.18	<u>39.74</u>	64.20
TR-DETR	V	63.30	47.22	62.89	42.12	41.53	39.69	62.80
OCS-DETR (Ours)	V	64.97	48.26	64.13	43.51	<u>43.03</u>	39.99	<u>63.01</u>
QD-DETR*	V + A	63.06	45.10	63.04	40.10	40.19	39.04	62.87
UVCOM	V + A	63.81	48.70	64.47	44.01	<u>43.27</u>	<u>39.79</u>	64.79
TR-DETR	V + A	65.05	47.67	64.87	42.98	43.10	39.90	<u>63.88</u>
OCS-DETR (Ours)	V + A	<u>64.13</u>	<u>48.06</u>	<u>64.57</u>	<u>43.97</u>	43.63	39.72	63.61

Table 1: The experimental results on the QVHighlights test set. The bold and underlined values represent the best result and the second-best result respectively. (*: Our baseline)

As shown in Table 1, we first evaluate OCS-DETR on QVHighlights, a benchmark designed for joint moment retrieval (MR) and highlight detection (HD) [3]. Compared with the strong DETR-based baseline QD-DETR [4], OCS-DETR consistently achieves better performance, especially on strict localization metrics. In the visual-only (V) setting, OCS-DETR reaches 64.97/48.26 on R@1@0.5/0.7, and improves mAP@0.75 from 39.88 to 43.51, with Avg. mAP increasing from 39.86 to 43.03. These gains indicate more accurate temporal boundary localization, which is critical for query-based moment understanding [3] [4].

For highlight detection, OCS-DETR achieves the best mAP (39.99) among visual-only methods under the “ $\geq$ Very Good” protocol, while maintaining competitive HIT@1 performance. This suggests that stronger moment-level alignment can also benefit clip-level saliency ranking in a unified framework [3].

When audio is incorporated (V+A), OCS-DETR further improves retrieval quality, achieving the best Avg. mAP (43.63) among compared methods, outperforming UVCOM [19] and TR-DETR [12]. Notably, the improvement remains most evident on mAP@0.75, indicating that

audio cues help refine temporal boundaries when combined with strong cross-modal alignment [3] [12] [19].

4.3.2 Results on TVSum Dataset

Performance comparison on the TVSum dataset.												
Method	Src	VT	VU	GA	MS	PK	PR	FM	BK	BT	DS	Avg
sLSTM	V	41.1	46.2	46.3	47.7	44.8	46.1	45.2	40.6	47.1	45.5	45.1
SG	V	42.3	47.2	47.5	48.9	45.6	47.3	46.4	41.7	48.3	46.6	46.2
LIM-S	V	55.9	42.9	61.2	54.0	60.3	47.5	43.2	66.3	69.1	62.6	56.3
Trailer	V	61.3	54.6	65.7	60.8	59.1	70.1	58.2	64.7	65.6	68.1	62.8
SL-Module	V	86.5	68.7	74.9	86.2	79.0	63.2	58.9	72.6	78.9	64.0	73.3
QD-DETR*	V	88.2	87.4	85.6	85.0	85.8	86.9	76.4	91.3	89.2	73.7	85.0
UniVTG	V	83.9	85.1	89.0	80.1	84.6	87.0	70.9	91.7	73.5	69.3	81.0
TR-DETR	V	<u>89.3</u>	93.0	94.3	85.1	88.0	88.6	80.4	<u>91.3</u>	89.5	<u>81.6</u>	88.1
OCS-DETR (Ours)	V	89.8	<u>90.7</u>	<u>93.6</u>	<u>85.2</u>	<u>87.4</u>	<u>88.3</u>	<u>78.9</u>	90.8	<u>88.9</u>	81.9	<u>87.5</u>
MINI-Net	V+A	80.6	68.3	78.2	81.8	78.1	65.8	75.8	75.0	80.2	65.5	73.2
TCG	V+A	85.0	71.4	81.9	78.6	80.2	75.5	71.6	77.3	78.6	68.1	76.8
Joint-VA	V+A	83.7	57.3	78.5	<u>86.1</u>	80.1	69.2	70.0	73.0	97.4	67.5	76.3
UMT	V+A	87.5	81.5	88.2	78.8	81.4	<u>87.0</u>	76.0	86.9	84.4	79.6	83.1
QD-DETR*	V+A	87.6	91.7	90.2	88.3	84.1	88.3	78.7	<u>91.2</u>	87.8	77.7	86.6
TR-DETR	V+A	90.6	92.4	91.7	81.3	86.9	85.5	<u>79.8</u>	93.4	88.3	<u>81.0</u>	<u>87.1</u>
OCS-DETR (Ours)	V+A	<u>89.4</u>	<u>92.1</u>	<u>90.9</u>	85.6	<u>86.1</u>	86.9	80.3	92.6	<u>88.7</u>	82.1	87.5

Table 2: The highlight detection results on TVSum. (*: Our baseline)

As shown in Table 2, we further evaluate highlight detection on TVSum [18], a widely used benchmark focusing on query-free video summarization. Following prior works, we report the average mAP score on TVSum as the main metric for highlight detection. In the visual-only setting, OCS-DETR achieves an average score of 87.5, clearly outperforming QD-DETR (85.0) [4], and approaching the strongest competing method TR-DETR [12]. This result demonstrates that the saliency modeling learned from joint MR–HD training can generalize well to a pure HD scenario [3] [4].

With visual and audio (V+A) inputs, OCS-DETR achieves the best overall average score (87.5) among all listed methods, surpassing both QD-DETR (86.6) [4] and TR-DETR (87.1) [12]. The consistent gains across categories suggest that OCS-DETR can effectively exploit audio information for highlight ranking without sacrificing stability, in line with observations from prior multimodal video understanding studies [12] [18].

4.3.3 Visualized Results

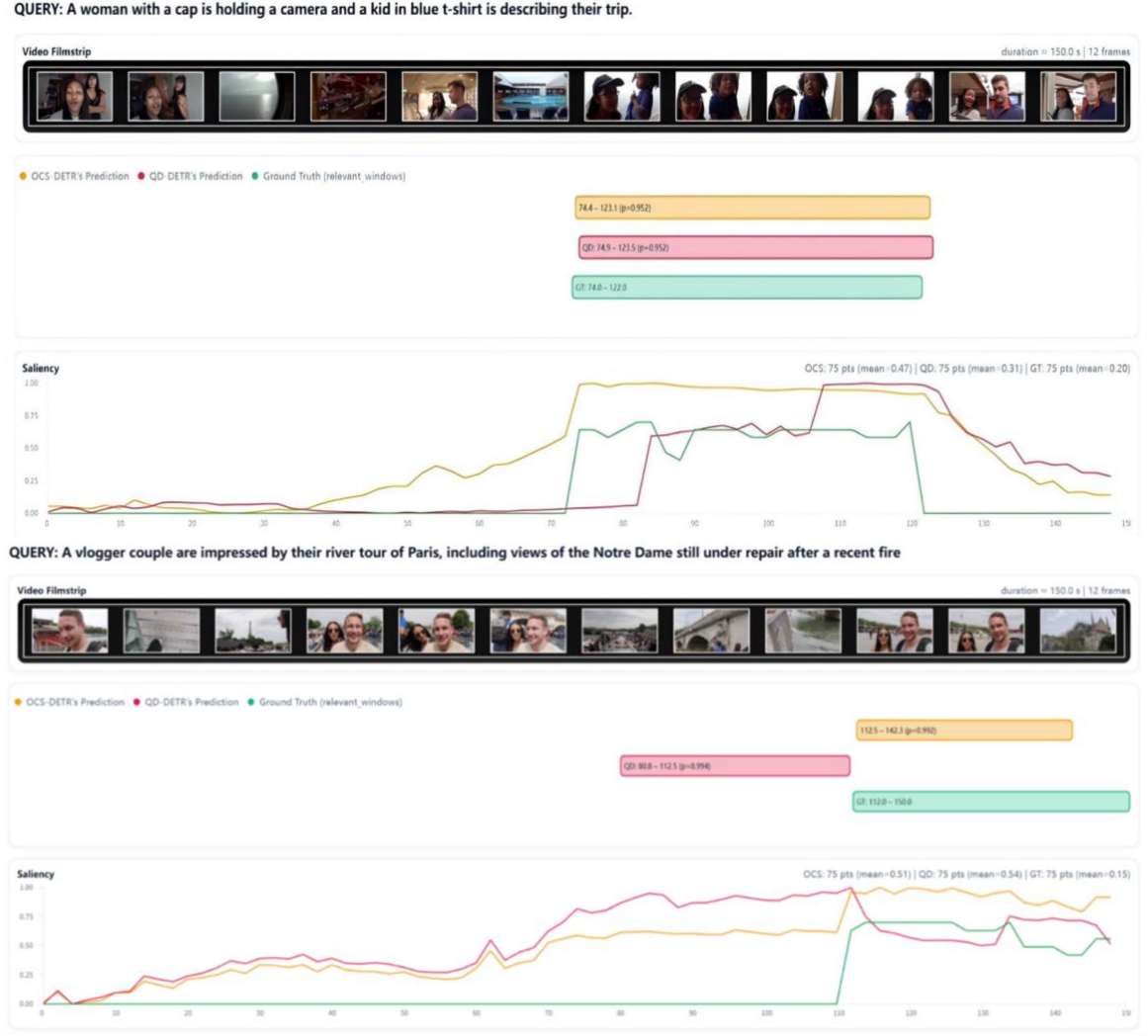


Figure 3: Qualitative comparison on QVHighlights

To better understand our model’s behavior, we visualize qualitative results of OCS-DETR on the QVHighlights validation set (Figure 3). Compared with the widely used baseline QD-DETR [4], OCS-DETR produces both tighter moment retrieval boundaries and more reliable highlight saliency patterns under the same query supervision provided by QVHighlights [3]. In the first example (camera + kid describing the trip), OCS-DETR predicts a moment that nearly overlaps the ground-truth window (OCS: 74.4–123.1 vs. GT: 74.0–122.0) and its saliency curve rises near the GT onset, stays high across the relevant interval, and decays after the GT ends, yielding a trend much closer to the GT saliency; in contrast, QD-DETR shows less consistent saliency with a late surge, which can lead to less stable scoring. In the second example (river tour in Paris and Notre Dame), OCS-DETR correctly focuses on the late part of the video (OCS: 112.5–142.3 vs. GT: 112.0–150.0), while QD-DETR is shifted earlier (80.8–112.5), missing most of

the relevant content; correspondingly, OCS-DETR’s saliency sharply increases at the GT onset and remains high afterward, whereas QD-DETR peaks before the GT begins and drops when the true relevant window starts. Overall, these cases demonstrate that OCS-DETR’s joint MR–HD modeling enables the two tasks to reinforce each other, producing saliency trajectories that better match the ground truth and, consequently, more stable and accurate temporal localization [3] [4].

4.4 Ablation Studies

Method	MR					HD					
	R1		mAP			Fair		Good		V. Good	
	@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1	mAP	HIT@1	mAP	HIT@1
Baseline	62.45	47.23	62.20	40.93	41.10	74.16	75.55	63.44	73.74	38.84	62.84
OT	63.61	47.23	63.29	41.48	41.14	74.22	74.90	63.29	72.52	39.06	61.94
CQA	63.61	46.39	62.88	41.97	41.50	75.03	75.42	64.08	74.00	39.41	63.10
Inter.	62.52	46.06	63.15	42.44	<u>42.13</u>	75.26	<u>76.84</u>	64.48	76.97	39.60	63.61
OT + CQA	62.71	<u>47.42</u>	62.84	<u>42.45</u>	41.62	74.76	75.61	63.84	74.06	39.39	<u>63.16</u>
OT + Inter.	63.02	46.84	63.42	42.32	41.73	74.46	75.73	63.96	74.37	39.49	63.09
CQA + Inter.	<u>64.45</u>	45.55	<u>63.80</u>	40.07	40.50	<u>75.75</u>	76.97	<u>64.83</u>	<u>74.65</u>	<u>39.85</u>	<u>63.16</u>
CQA + Inter. + OT	64.97	48.26	64.13	43.51	43.03	76.07	76.13	64.95	74.06	39.99	63.01

Table 3: Ablation study results on MR and HD after removing component

As shown in Tabel 3, this ablation study on QVHighlights shows that each component contributes differently to MR and HD, and the best performance is obtained when they are combined. Starting from the baseline (Avg. mAP = 41.10), adding OT alone yields only marginal improvement (Avg. mAP = 41.14), indicating limited benefit when used independently. CQA provides a clearer gain, improving both MR (Avg. mAP = 41.50) and HD (V.Good mAP 38.84 → 39.41). The most critical factor is Inter., which significantly boosts strict localization for MR (mAP@0.75 40.93 → 42.44; Avg. mAP 42.13) and also improves HD ranking/selection (Good HIT@1 73.74 → 76.97; V.Good HIT@1 62.84 → 63.61), validating that MR–HD interaction yields more stable and discriminative temporal reasoning. While CQA + Inter. improves R@1@0.5 and achieves strong HD performance (V.Good mAP = 39.85), it drops on mAP@0.75 (40.07), suggesting the strict-boundary refinement is still insufficient. Introducing OT on top of CQA + Inter. resolves this issue and leads to the overall best results: the full model reaches R@1@0.5/0.7 = 64.97/48.26, mAP@0.75 = 43.51, and Avg. mAP = 43.03, while also achieving the best V.Good mAP (39.99) for HD. Overall, the results confirm the complementarity of alignment (CQA), task interaction (Inter.), and auxiliary optimization (OT), which together produce consistent gains across both MR and HD metrics.

5 Conclusion

This work proposes OCS-DETR for joint moment retrieval (MR) and highlight detection (HD), aiming to improve cross-modal grounding and strict boundary localization under the same DETR-style framework [2] [4]. To ensure fair comparison, we inherit the same frozen feature extraction pipeline from QD-DETR, joining CLIP visual features with SlowFast motion features for video, optionally adding PANN audio features, and using CLIP text features for queries; therefore, the performance gains mainly come from our alignment and fusion design rather than stronger backbones [4].

Specifically, we introduce an OT-based alignment module using Sinkhorn to better match query tokens and video clips and reduce noisy cross-modal associations [5]. We further design a Visual Refinement Fuser (VRFuser) based on context–query attention and concatenation to suppress query-irrelevant video content and produce more discriminative joint representations [8] [9]. In addition, we propose an MR–HD interdependency module where saliency-guided refinement improves temporal reasoning and stabilizes strict localization. Finally, for the MR objective, we replace the original CE classification with sigmoid focal loss to address extreme foreground–background imbalance and emphasize hard negatives, while keeping the HD supervision consistent with QD-DETR (margin + rank) for fair evaluation [4] [6].

Extensive experiments on QVHighlights and TVSum verify the effectiveness of OCS-DETR. OCS-DETR consistently improves MR, especially on strict metrics such as mAP@0.75, and achieves competitive or best HD performance under both visual-only and audio-augmented settings [3] [4]. In the future, we will explore more efficient and principled multi-modal interaction mechanisms to better coordinate audio–visual–text cues and further enhance robustness on long, noisy videos.

Method	Src	Moment Retrieval					Highlight Detection	
		R1		mAP			$\geq$ Very Good	
		@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1
Base + Direct Audio	V + A	64.13	48.06	64.57	43.97	43.63	39.72	63.61
Base + Gated Audio	V + A	63.42	48.71	64.01	43.20	42.69	39.23	61.81
Base + Video OT Audio	V + A	64.19	48.84	64.30	43.37	43.11	39.82	62.13
Base + Video OT + Saliency Audio	V + A	62.19	46.13	62.74	40.92	40.89	39.57	62.77

Table 4: Attempt on audio

The results in Table 4 demonstrate that the method of audio integration is crucial: direct audio fusion achieves outstanding joint MR/HD performance (Avg. mAP of 43.63), whereas other designs such as gated audio or OT-based audio variants do not consistently improve both moment retrieval and highlight ranking accuracy simultaneously, and may even degrade certain metrics (HD mAP/HIT@1). Consequently, future research will focus on more robust audio modeling and better alignment of audio events with highlights to further enhance the framework's multimodal capabilities.

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Appendices

Our Other Works

Method	Performance Metrics										
	MR					HD					
	R1		mAP			Fair		Good		V. Good	
	@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1	mAP	HIT@1	mAP	HIT@1
Rotary	62.45	47.23	62.2	40.93	41.1	74.16	75.55	63.44	73.74	38.84	62.84
Blip	60.39	44.45	61.65	40.63	40.27	73.22	73.16	62.28	71.68	38.37	60.71
GRAM	63.68	47.23	62.2	40.93	41.1	74.16	75.55	63.44	73.74	38.84	62.84

Figure A1: We conducted these auxiliary trials to test whether off-the-shelf upgrades could close the performance gap without changing the main architecture: (i) replacing CLIP visual features with BLIP to strengthen vision–language semantics, (ii) injecting RoPE position embedding to video features to better encode temporal order in clip sequences, and (iii) using a GRAM-style alignment to improve cross-modal matching. However, all variants perform worse than the QD-DETR baseline, so we do not adopt them in the final OCS-DETR design [4] [20] [21][22].

Besides our base works, we have tried more methods to improve our modal performance (as shown in Figure A1):

Given that our baseline still relies on frozen feature extractors and standard temporal encoding, we hypothesize that stronger vision–language pretraining (BLIP [20]), improved sequence position modeling (RoPE [21]), or alternative feature alignment (GRAM [22]) might yield a free performance boost without touching our core OCS modules.

The BLIP variant shows the most consistent drop: MR $R1@0.7$ decreases to 44.45 (vs. 47.23 with RoPE), MR mAP Avg becomes 40.27 (vs. 41.10), and HD HIT@1 falls from 75.55/73.74/62.84 (Fair/Good/V.Good under RoPE) to 73.16/71.68/60.71. RoPE and GRAM are closer to each other—GRAM slightly raises MR $R1@0.5$ to 63.68 (vs. 62.45), but keeps MR mAP Avg at 41.10 and yields identical HD scores (74.16/63.44/38.84 mAP for Fair/Good/V.Good), indicating limited practical benefit. Overall, these results suggest that “backbone swap (CLIP to BLIP)” or “positional tweak (RoPE)” alone does not resolve the core issue in joint MR&HD, i.e., noisy cross-modal association; the improvements of OCS-DETR mainly come from our alignment and fusion design rather than stronger feature extractors or generic sequence priors [4].